香港中文大學(深圳)
The Chinese University of Hong Kong, Shenzhen

PHY1001 Mechanics

2023-2024 Term 2

Final Examination

May 12th, 2024; Time Allowed: 3 Hours

NAME (print) _____

CUHKSZ ID _____

Room No. _____

Seat No. _____

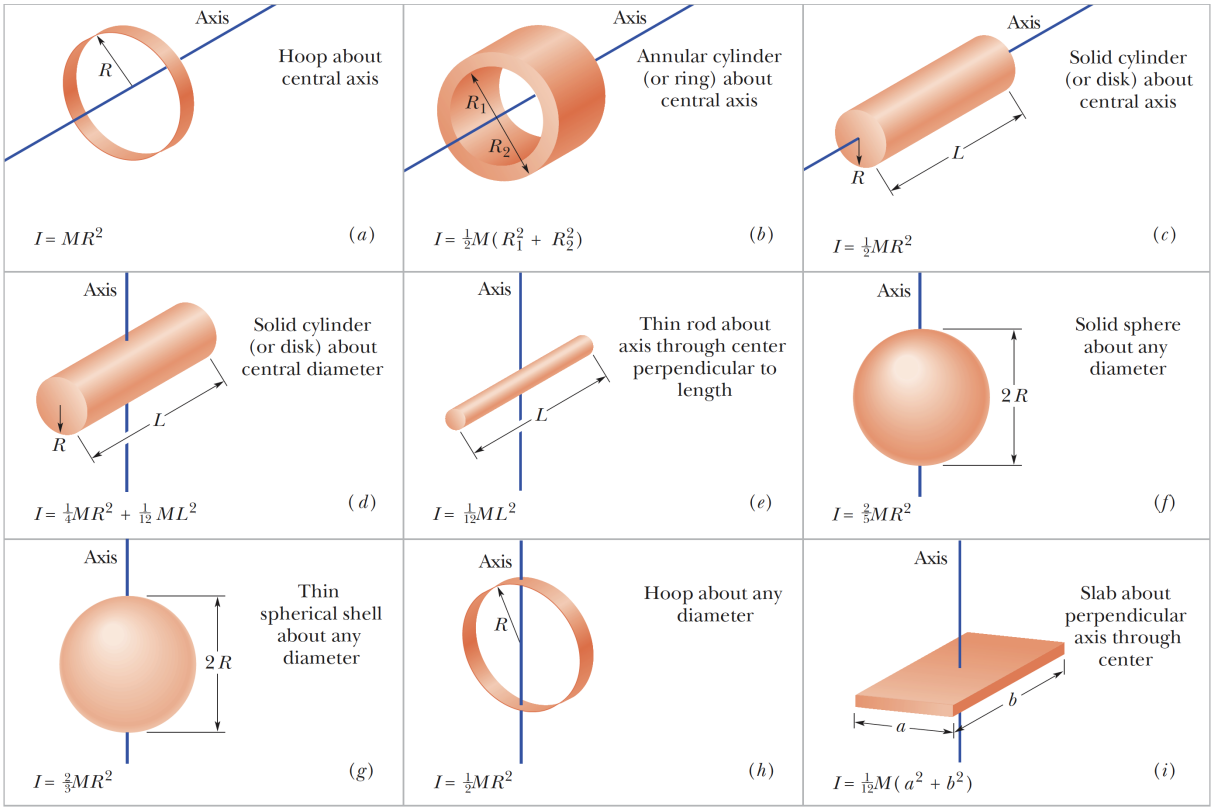
- **Show all your work**. Correct answers with little supporting work will not be given credit.
- Closed Book Exam: One piece of double-sided A4 reference paper, a scientific calculator, and a paper-based dictionary are allowed.
- Unless approved by the instructors, students who arrive more than 30 minutes late will NOT be admitted.
- The total points are 120 points. You need to finish ALL the questions in 3 hours (180 minutes).

#1	#2	#3	#4	#5	#6	#7	#8	#9	#10	#11	#12	TOTAL
10	10	10	10	10	10	10	10	10	10	10	10	120

Summary of Basic Calculus:

$$\frac{d}{dx}x^n = nx^{n-1},$$
$$\frac{d}{dx}e^{ax} = ae^{ax},$$
$$\frac{d}{dx}\cos ax = -a\sin ax, \quad \frac{d}{dx}\sin ax = a\cos ax,$$
$$\frac{d}{dx}\ln ax = \frac{1}{x},$$
$$\frac{d}{dx}(uv) = v\frac{d}{dx}u + u\frac{d}{dx}v,$$
$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \ (n \neq -1),$$
$$\int \frac{dx}{x} = \ln x + C,$$
$$\int e^{ax}dx = \frac{1}{a}e^{ax} + C, \quad C \text{ is a constant.}$$

Table of Moments of Inertia:



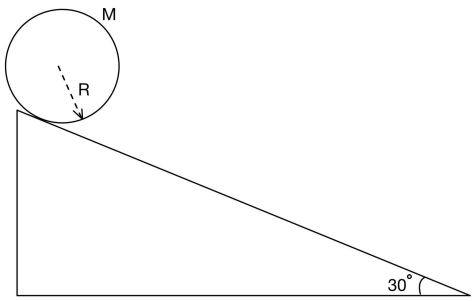
Bernoulli's Equation

$$P + \rho g y + \frac{1}{2}\rho v^2 = \text{constant.}$$

Spring Mass System; Simple Harmonic Oscillator

$$m\frac{d^2x}{dt^2} = -kx.$$

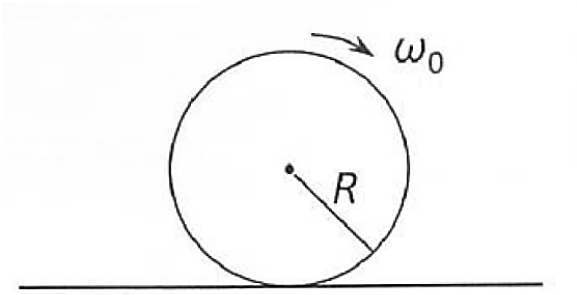
1. A uniform cylinder of mass M and radius R moves down a 30° ramp from stationary. **(10 pts)**



(a) Find the magnitude of the linear acceleration of the cylinder in smooth rolling (rolling without slipping). ($g = 9.80\,\mathrm{m/s^2}$) **(6 pts)**

(b) If the cylinder rolls down the ramp smoothly without any slipping, find the minimum static frictional coefficient. **(4 pts)**

2. A uniform solid cylinder with mass M and radius R is given an initial clockwise angular velocity ω_0 and zero linear velocity ($v_{com} = 0$) before it is placed on a horizontal surface with kinetic friction coefficient μ_k. **(10 pts)**



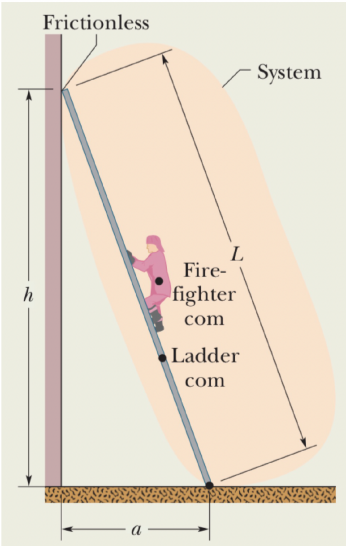
(a) Which direction is the cylinder going to move towards? (Hint: Left or right) **(2 pts)**

(b) What is the acceleration at the moment that the cylinder is placed on the surface? Please indicate both the magnitude and direction. **(2 pts)**

(c) Find that the corresponding angular acceleration of the cylinder. Please indicate both the magnitude and direction. **(2 pts)**

(d) How long does it take for the cylinder to roll without slipping (smooth rolling)? (Hint: The rolling without slipping happens when $\omega R + v = 0$) **(4 pts)**

3. As shown in the figure below, ladder of length $L = 10.0\,\text{m}$ and mass $m = 50.0\,\text{kg}$ leans against a slick (frictionless) wall. Its upper end is at height $h = 8.60\,\text{m}$ above the floor on which the lower end rests (the floor is not frictionless). The ladder's center of mass is $L/3$ from the lower end. A firefighter of mass $M = 75.0\,\text{kg}$ climbs the ladder until her center of mass is $L/2$ from the lower end. $g = 9.80\,\text{m/s}^2$. **(10 pts)**

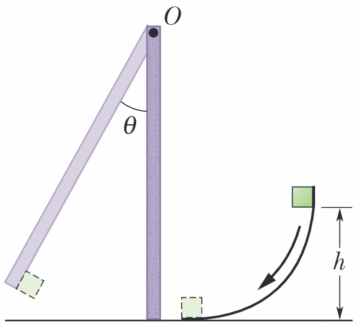


(a) What is the normal force acting from the wall on the ladder? Please indicate both the magnitude and direction. **(4 pts)**

(b) What is the frictional force acting from the floor on the ladder? Please indicate both the magnitude and direction. **(3 pts)**

(c) What is the normal force acting from the floor on the ladder? Please indicate both the magnitude and direction. **(3 pts)**

4. In the figure below, a small block with $m = 5.00 \times 10^{-2}$ kg slides down a frictionless surface through height $h = 1.00 \times 10^{-1}$ m and then sticks to a uniform rod of mass $M = 1.50 \times 10^{-1}$ kg and length $l = 3.50 \times 10^{-1}$ m. The rod pivots about point O through angle θ before momentarily stopping. $g = 9.80\,\mathrm{m/s^2}$. **(10 pts)**



(a) Find the speed of the block before collision. **(2 pts)**

(b) Find the angular speed of the block-rod system immediately after the collision. **(3 pts)**

(c) Find the value of $\cos\theta$ with θ the maximum angle. **(5 pts)**

5. A rock is suspended by a light string. When the rock is suspended in the air at rest, the tension in the string is 39.6 N. When the rock is totally immersed in water, the tension is 28.4 N. (Hint: The density of water is $1.00\,\mathrm{g/cm^3}$. Neglect the buoyancy of the air. $g = 9.80\,\mathrm{m/s^2}$) **(10 pts)**

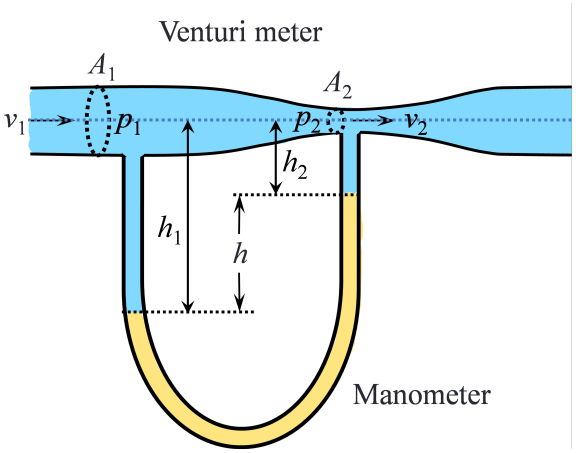
   (a) What is the buoyancy of water on the rock? Indicate both the magnitude and direction. **(2 pts)**

   (b) What is the reaction force of the rock against the water? Indicate both the magnitude and direction. **(2 pt)**

   (c) What is the volume of the rock? (Hint: Keep 3 significant figures.) **(3 pts)**

   (d) What is the density of the rock? (Hint: Keep 3 significant figures.) **(3 pts)**

6. As shown below, the Venturi meter is connected to a U-shaped manometer. Consider an ideal fluid of density ρ_1 flowing through the horizontal pipe. The fluid at the portion of the pipe with a larger cross-section area A_1 has a flow speed v_1 and a measured pressure of p_1. The fluid at the portion of the pipe with a narrower cross-section area A_2 has a flow speed v_2 and a measured pressure of p_2. There is a steady flow in the horizontal pipe and the height difference of the fluid in the manometer remains constant at $h = h_1 - h_2$. **(10 pts)**



(a) Write down the continuity equation for the horizontal pipe. **(2 pts)**

(b) Use Bernoulli's equation to show $v_1 = \sqrt{\dfrac{2(p_1 - p_2)}{\rho_1[(A_1/A_2)^2 - 1]}}$ **(4 pts)**

(c) Consider two different fluids of density ρ_1 (fluid in the horizontal pipe) and ρ_2 (fluid in the U-shaped manometer), where $\rho_1 < \rho_2$. Determine the flow speed v_1 in terms of ρ_1, ρ_2, A_1, A_2 and h, using part b. **(4 pts)**

7. A piano wire with mass $m = 4.00 \times 10^{-3}$ kg and length $l = 0.800$ m is stretched. A sinusoidal wave travels along the wire with frequency $f = 2.50 \times 10^2$ Hz, wavelength $\lambda = 0.120$ m, and amplitude $A = 2.40 \times 10^{-3}$ m. **(10 pts)**

(a) Find the wave number k. **(1 pt)**

(b) Find the angular frequency ω. **(1 pt)**

(c) Find the period T. **(1 pt)**

(d) Find the speed v of wave propagation. **(1 pt)**

(e) Find the tension F_T (or τ) in the wire. **(2 pts)**

(f) The average power carried by the wave is $p_{av} = \frac{1}{2}\sqrt{\mu F_T}\omega^2 A^2$. Find the work W required to generate the wave for one period. **(2 pts)**

(g) If the frequency f is increased by a factor of 2 to $f' = 5.00 \times 10^2$ Hz. Find the work W' required to generate the wave for one period. **(2 pts)**

8. The following two functions describe two waves:

$y_1 = (2.0 \times 10^{-3}\,\mathrm{m})\sin[(5.0\mathrm{m}^{-1})x - (45\mathrm{s}^{-1})t]$,
$y_2 = (2.0 \times 10^{-3}\,\mathrm{m})\sin[(5.0\mathrm{m}^{-1})x + (45\mathrm{s}^{-1})t]$.

(a) Choose the wave propagation direction. **(1 pt)**

y_1: [___] +x direction, [___] -x direction.
y_2: [___] +x direction, [___] -x direction.

(b) The superposition of the two waves should be a standing wave. Write down the wave function y of the standing wave. (Hint: $\sin A + \sin B = 2\sin(\frac{A+B}{2})\cos(\frac{A-B}{2})$.) **(1 pt)**

(c) What is the maximum displacement $y_{\max}$ of any point on the standing wave? **(1 pt)**

(d) Antinodes of the standing wave can achieve the maximum upward or downward displacement. Find the antinode positions. (Hint: express the positions with integer n.) **(2 pts)**

(e) Define $y = 0$ at all time as the requirements of nodes of the standing wave. Find the node positions. (Hint: express the positions with integer n.) **(2 pts)**

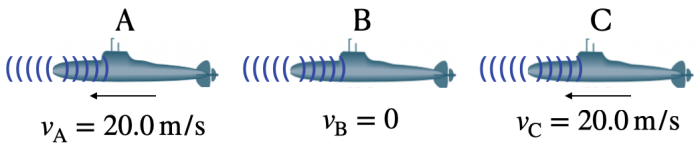
(f) Particle oscillation speed v_y(or u) $= \dfrac{\partial y}{\partial t}$. Write down the speed function v_y of particles on the standing wave. **(1 pt)**

(g) What is the oscillation amplitude of particle velocity as a function of x? **(1 pt)**

(h) Where is the maximum amplitude of the oscillation velocity? **(1 pt)**

9. Doppler effect: $f' = \dfrac{v \pm v_D}{v \pm v_S} f$. v_D is the speed of detector and v_S is the speed of sound source. v is the speed of sound.

As shown in the figure below, three submarines are at the same depth in <u>motionless</u> water. Sub A moves to the left at speed $v_A = 20.0\,\text{m/s}$, sub B is at rest, and sub C moves to the left at speed $v_C = 20.0\,\text{m/s}$. All three submarines send sonar signals at frequency $f = 2.82 \times 10^3\,\text{Hz}$. Sonar waves travel at $v = 1.52 \times 10^3\,\text{m/s}$ in water. **(10 pts)**



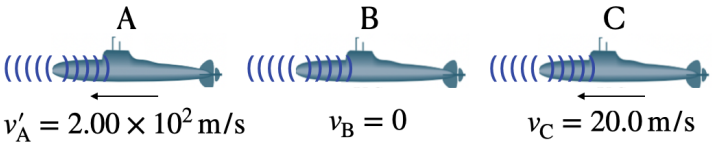
(a) What is the frequency f_{AB} detected by sub B from sub A? **(2 pts)**

(b) What is the frequency f_{BC} detected by sub C from sub B? **(2 pts)**

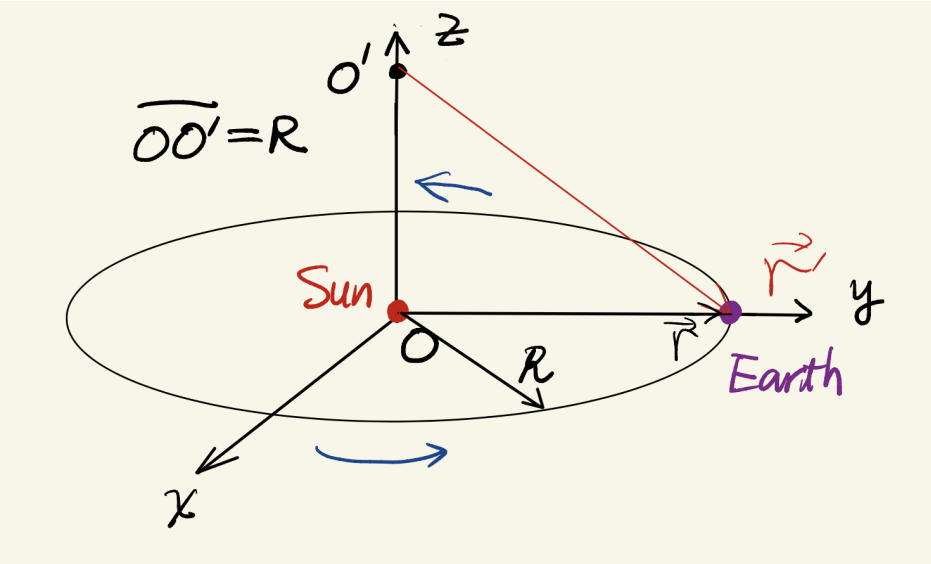
(c) What is the frequency f_{AC} detected by sub C from sub A? **(2 pts)**

(d) What is the frequency f_{BA} detected by sub A from sub B? **(2 pts)**

(e) Sub A gets black technology from aliens and increases its speed to $v'_A = 2.00 \times 10^2\,\text{m/s}$, as shown in the figure below. What are the frequencies f'_{AB} (detected by sub B from sub A) and f'_{BA} (detected by sub A from sub B)? **(2 pts)**



10. Orbital Angular Momentum of the Earth **(10 pts)**



Consider the Sun-Earth system with masses M_S and M_E, assuming $M_s \gg M_E$ and neglecting other planets and moons. Treat the Earth as a point particle and its orbit as a circular orbit with a radius of R. In the diagram above, the Sun is located at point O, while point O' is directly above O at a distance of R. At one certain moment, the earth is on the y axis, and moving toward the $-x$ direction. (You may refer to this position when you solve for part (c) and part (d)).

(a) Find the period T_0 it takes for the earth to finish one turn (circle). **(2 pts)**

(b) Choose O as the reference point and compute the orbital angular momentum $\vec{L}_0$ of the earth. Hint: Provide both the magnitude and direction according to $\vec{L}_0 = M_E \vec{r} \times \vec{v}$. **(2 pts)**

(c) Now choose O' as the reference point, and compute the orbital angular momentum $\vec{L}$ of the earth. Hint: $\vec{L}$ is different from $\vec{L}_0$. Give the magnitude and draw the direction. **(2 pts)**

(d) Compute the torque $\vec{\tau} \equiv \vec{r}' \times \vec{F}$ about the reference point O' generated by the gravitational pull (force) of the Sun on the Earth ($\vec{F}$). **(2 pts)**
Hint: Give the magnitude and direction!

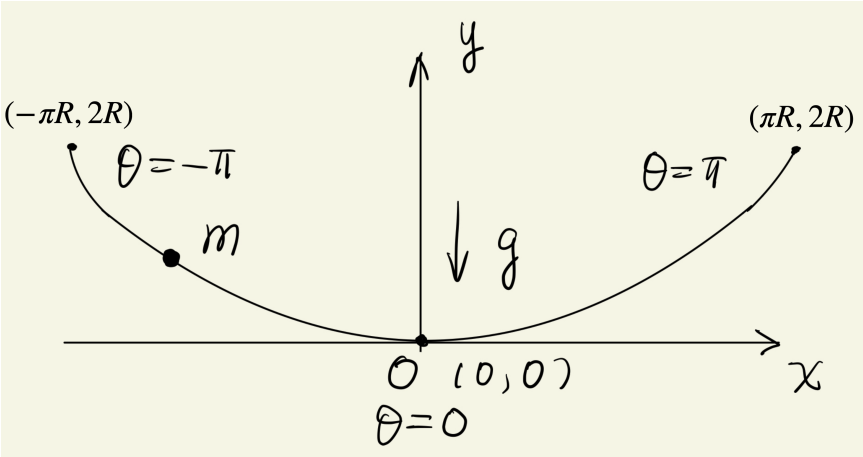
(e) Choose O' as the reference point, then use Newton's second law for rotation $\vec{\tau} = \dfrac{d\vec{L}}{dt}$ to explain the change of the orbital angular momentum L. Compute the angular frequency Ω of this "rotation" of the orbital angular momentum. **(2 pts)**

11. Oscillation on the Cycloidal Wire. **(10 pts)**

A bead of mass m moves on a smooth (frictionless) wire shaped as a cycloid:

$$x = R(\theta + \sin\theta), \qquad y = R(1 - \cos\theta),$$

where R is a positive constant and the parameter θ lies in the range $-\pi \le \theta \le \pi$.



The wire is fixed at point O, with the y-axis pointing vertically upwards and the x-axis pointing horizontally. As shown in the figure above, the gravitational force acts in the $-y$ direction. Since the cycloidal wire can be parametrized by θ, one can describe the motion of the bead on the wire in terms of $\theta(t)$, which is a function of time t.

(a) Find the x and y components of the velocity of the bead. **(2 pts)**

Hint: $v_x = \dfrac{dx}{dt}$ and $v_y = \dfrac{dy}{dt}$. Express your results in term of θ and the time derivative of θ, which is given as $\dot{\theta} \equiv \dfrac{d\theta}{dt}$.

(b) Find the kinetic energy of the bead sliding on the wire. **(1 pt)**

(c) Consider the gravitational potential together with the kinetic energy, and show that the mechanical energy of the bead sliding on the wire

$$E = mR^2\dot{\theta}^2\,[1+\cos\theta] + mgR(1-\cos\theta)$$

is a constant. **(1 pt)**

(d) By changing variable from θ to $u \equiv \sin\dfrac{\theta}{2}$, rewrite the expression for E from Part (c). Hint: Express your results in terms of the new variables u and $\dot{u} \equiv \dfrac{du}{dt}$. **(2 pts)**
You may find these two trigonometric identities useful:

$$\sin^2\frac{\theta}{2} = \frac{1-\cos\theta}{2} \quad \text{and} \quad \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}.$$

(e) By noting that E is a constant and differentiating $E(u, \dot{u})$ with respect to t, derive the equation of motion for the bead. **(2 pts)**

(f) Based on the above equation of motion, is the oscillation of the bead about $x = 0$ a simple harmonic motion? (Yes/No) **(1 pt)**

(g) Find the angular frequency ω for the bead oscillating on the cycloidal wire. **(1 pt)**
You do not have to solve the equation of motion. Instead, you can use an analogy with the spring-mass system, which has the following equation of motion and angular frequency

$$\frac{d^2x}{dt^2} + \frac{k}{m}x = 0, \quad \text{and} \quad \omega = \sqrt{\frac{k}{m}}.$$

12. The Restricted Three-Body Problem: **(10 pts)**

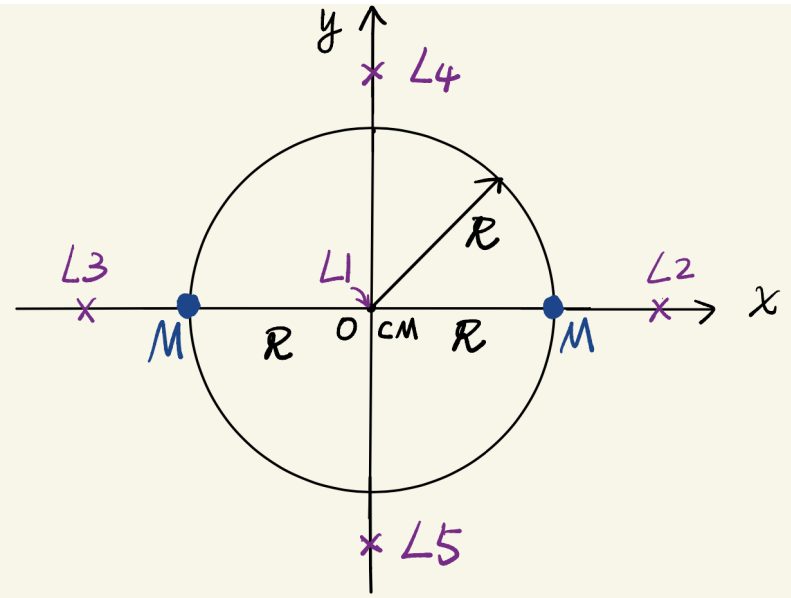
In a galaxy far, far away, the legendary Starship Enterprise discovered a new two-star system: Two equal-mass stars, each with mass M, are rotating about their common center of mass in a circular orbit with radius R.

Captain Kirk, renowned for his decisive actions, commands Enterprise to park at a Lagrange point of the two-star system with minimal fuel consumption. At the Lagrange points, the gravitational forces of the two large bodies provide the centripetal force, allowing the Starship to orbit with the same angular velocity as the stars. These three bodies remain in a co-moving configuration, and the Enterprise does not have to use its power.

Your task is to determine the locations of the Lagrange points where Enterprise can be parked under the following three assumptions

- Neglect the gravitational effects of the Starship with mass m on the stars since $M \gg m$.
- Consider the system to be in a two-dimensional plane.
- Assume the stars and the Starship move in circular orbits.

There are five Lagrange points in total. In this special case with two equal mass stars, the first point L_1 locates at the center of mass of these two stars. In the following, we can find the rest of the Lagrange points L_2 - L_5.



(a) Find the magnitude of the angular velocity of the two-star system. **(2 pts)**

Hint: these two stars are circulating about their CM under the gravitational force $GM^2/(2R)^2$.

(b) Find the equation that determines the locations of L_2 and L_3, which lie on the line through the two stars. **(3 pts)**

(Don't Panic! You do not have to solve this equation during the exam.)

(c) Find the locations (the y coordinates) of the L_4 and L_5, which are on the line perpendicular to the line through the two stars. **(3 pts)**.

(d) Now consider the case that these two stars have different masses such as M_1 and M_2 and assume the distance between these two stars is still $2R$. Find the magnitude of the angular velocity again by repeating the calculation in Part (a). **(1 pt)**

(e) Following part (d), will the locations of L_4 and L_5 change as compared to the result in part (c)? (Yes/No) **(1 pt)**

Hint: You do not have to give a reason. A Yes/No answer is sufficient.